

Experiment#03

Objective:

To perform Boolean Expression of ($Y = ABC + \bar{A}BC + A\bar{B}C$).

Equipment's:

- Breadboard
- Jumper wires
- Power supply
- Integrated ICs
- LED light

Theory:

Boolean Expression:

- Boolean algebra is a type of algebra that operates within the binary system (using only two values: 0 and 1).
- It was proposed by the English mathematician George Boole in 1854.
- This algebra is based on Aristotle's propositional logic, using symbols for True (1) and False (0).
- Boolean algebra deals with binary variables and logic operations.
- It plays a fundamental role in the development of digital electronics systems and also finds applications in Set Theory, Statistics, and other branches of mathematics.

Basic Boolean Operations

1. Negation (NOT) Operation:

- Reverses the value of a Boolean variable from 0 to 1 or vice versa.
- If $A = 1$, then $\text{NOT}(A) = 0$.
- If $A = 0$, then $\text{NOT}(A) = 1$.
- We represent negation as $\sim A$ (e.g., if $A = 1$, $\sim A = 0$).

2. Conjunction (AND) Operation:

- Satisfies the condition if both individual variables are true.
- If $A = \text{True}$ and $B = \text{True}$, then $A \text{ AND } B = \text{True}$.
- If $A = \text{True}$ and $B = \text{False}$ (or vice versa), then $A \text{ AND } B = \text{False}$.
- If both A and B are false, $A \text{ AND } B = \text{False}$.

3. Disjunction (OR) Operation:

- Satisfies the condition if any individual variable is true.
- If $A = \text{True}$ and $B = \text{True}$, then $A \text{ OR } B = \text{True}$.

- If A = True and B = False (or vice versa), then A OR B = True.
- If both A and B are false, A OR B = False.

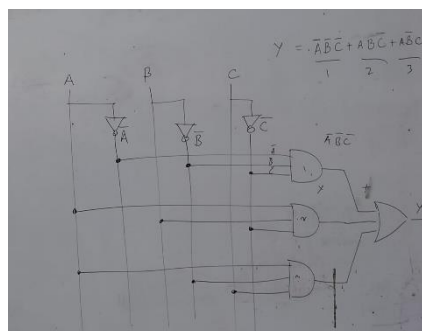
Boolean Expressions and Variables

- A Boolean expression produces a Boolean value when evaluated (either true or false).
- Boolean variables store Boolean numbers (0 or 1).
- For example, the expression $P + Q = R$ involves Boolean variables P, Q, and R, each capable of storing values 0 or 1.

Procedure:

- First, collect all the required equipment and components that are used.
- Now make a circuit diagram according to the given equation.
- Make a circuit on the breadboard according to the circuit diagram.
- Put LED/DMM to check our circuit
- After making the circuit give a power supply to the circuit with the help of DC power supply and jumper wires.
- Check all the connections have been connected according circuit diagram.
- Now give supply to the circuit to make the truth table by changing values.

Circuit diagram:



Fig#01

Picture:



Fig#02

Truth table:

A	B	C	A'	B'	C'	ABC	ABC	ABC	Y=ABC+ABC+ABC

Conclusion:

In this experiment, we have learned how to make a circuit diagram with the help of equations and to make a circuit with the help of ICs on the breadboard. This experiment also helps to learn how we connect different ICs to our desired output.